Wearable firmware: the Singing Jacket

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Abstract

This paper discusses a design for a purpose-built interface that a wearer uses for controlling sound through movement. It presents work done so far on a mobile sound installation called the Singing Jacket and looks at ways in which the design can benefit from new developments in microcontroller technology. Several different microcontrollers are reviewed along with additional peripherals used for sensing movement and creating sound. These developments are presented against a backdrop of problems affecting musicians whose creative output is intimately bound up with changes that have been brought about by the ongoing development of electronic technology.

1 Electronic technology and disposable music

The last half of the twentieth century witnessed an unprecedented transformation in the way music is created, produced and distributed. Much of this change can be attributed to the revolution in electronics. The way musicians work has been transformed by changing technology and differs fundamentally from the way that musicians worked a century earlier. Electronic music composition involves new performance paradigms and frequently crosses disciplinary boundaries; its composers often perform their own music and a growing number are directly involved in the conception and implementation of new instrument designs and systems used for their creative work.

Electronic music composition poses a special problem. In the case of electronic composition, subsequent performance and reinterpretation is no longer possible when the technology associated with a particular work becomes superceded. This problem does not affect works composed using more conventional musical resources to quite the same extent.

Herein lies the paradox. Works by composers who have been proactive in the development of new forms of electronic music have become, by virtue of the accelerated rate of technological change, isolated from the very mainstream these works helped to bring about. Many of the early works of electronic music have become trapped within a period of technological development that has been superseded by more advanced technologies.

This phenomenon affects all composers of electronic music alike, both those engaging directly in building new electronic instruments as an integral part of the compositional process, and those working with new technology along more traditional lines.

Some examples of this phenomenon include music composed by Don Banks for ensembles that included keyboard instrument like the VCS-3 or Quasar M8 or theremin-based performances created by the Phillipa Cullen Dance ensemble in the early 1970s [7] or audiovisual environmental pieces works written by Martin Wesley-Smith for the Fairlight CMI [5].

There are two possible responses to this paradox:

1.1 The safe approach will be to focus all efforts on new musical activity using standard electronic musical instruments developed by instrument manufacturers. It would seem logical to argue that it is now possible to address this phenomenon by using instruments that have been engineered for the broadest cross section of musicians.

The problem is that by the time the product has been developed and matured to a point where it is useable by musicians in general, the technology on which it is based has been overtaken. The culprit is none other than the ongoing development of faster processors, new operating systems and, more importantly, new performance paradigms that grew out of earlier generation of technology.

1.2 The alternative response is more reckless: the musician develops generic technology and takes an active role in the ongoing development of non-standard instruments that appropriate consumer products manufactured for non-musical purposes. This approach does not sit easily with the musical constraints imposed by manufacturers of electronic musical instruments.

There are many precedents for musicians who take this approach. The new performance paradigms that continue to destabilise the technology on which standard electronic music is founded are a legacy of an earlier generation of musicians whose creative work embraced technological issues and laid the foundations of electronic music. The work of these musicians is documented elsewhere [18].

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2. Firmware design environments

Microcontroller technology has been responsible for much of the transformation that has taken place in music over the past decade and a half. Microcontrollers can be found in point-of-sales terminals, vending machines, domestic appliances, consumer goods, communications equipment, mobile phones and in engines and tyres of motor vehicles. Many are as powerful as the first computer music systems in the early 60s.

Microcontroller applications operate on firmware i.e. a dedicated software application that resides in non-volatile memory and runs automatically when power is applied. Firmware offers the advantage of a dedicated application that will run without the constraints of the disk operating system and the graphic user interface, or GUI.

While development of new composition systems has been focused on the GUI (the legacy of desktop computing) so far little has been done to create composition tools that focus on the interface between computer technology and the real world.

Notable exceptions to this include the MIDI Tool Box which uses a HC11 microcontroller [18][19], STEIM's SensorLab [21] and the Smart Controller [8] which uses field programmable gate array logic (GAL). While GAL is not the same as firmware in the strictest sense (i.e. conventional executable program instructions), it should be included in a summary of approaches attempting to address deficiencies in the user interface that standard composition environments in the desktop environment alone do not address.

2.1 The Singing Jacket

An example of a recent musical development using microcontroller technology is the *Singing Jacket* developed by Mark Havryliv.

The Singing Jacket is a mobile sound installation that is worn by a listener. It produces pre-composed sound that is influenced by the wearer's movement. The work creates a personal sound space within a larger acoustic environment.

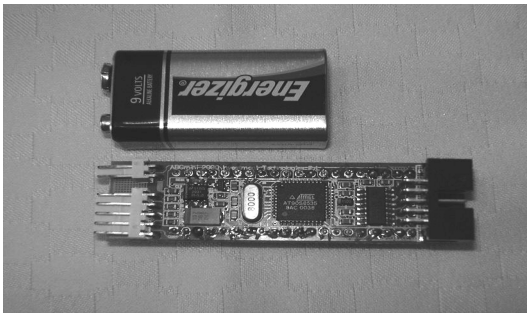


Figure 1.AVR AT90S8535 microcontroller

At the core of the Jacket is a small, power efficient microcontroller (AVR AT90S8535) which is programmed to produce sound that is influenced by a wearer's movement. The microcontroller is shown in Figure 1. The microcontroller has non-volatile memory on-chip which is used to store the program. The program determines the complexity of the interaction between the wearer's movement and the sound produced.

The first application for the Jacket generates sound by toggling (i.e. alternately switching between logic 1 and 0) adjacent output pins of the microcontroller. The pins are connected directly to a series of piezo-ceramic transducers to produce square waves.

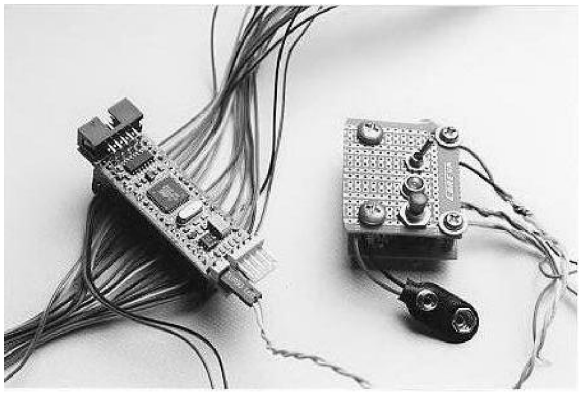


Figure 2. Cable harness connects microcontroller to transducers and switches (not shown)

Sound is output from 24 transducers. All frequencies produced are subdivisions of the processor clock, thereby producing ratios of the subharmonic series. Movement is detected using 8 mercury tilt switches that are sewn onto the fabric of the jacket arms.

The Singing Jacket is a departure from both the MIDI Tool Box and the Smart Controller in so far as it represents a recent paradigm shift that has taken place within the technology itself. Whereas the MTB uses a microcontroller designed for controlling motor vehicles, the Jacket uses AVR technology designed for battery-powered handheld appliances. Unlike the Smart Controller which is programmed using a graphic user interface the Jacket, like the MIDI Tool Box, is programmed entirely using low level text-based tools.

Further developments are planned for the Jacket. These will see the jacket become part of a development environment that is both wearable, self contained and able to connect to the outside world. Wearable implies a different level of mobility and power consumption than that associated with the first microcontrollers. It also implies a new kind of technology that is indistinguishable from everyday apparel [16].

3. Development issues - what will I wear tomorrow?

The kind of interactive works suggested by version 1 of the Jacket provided a point reference as we tried to identify the most appropriate choice of microcontroller interface.

In order to pursue these developments further, we asked two questions:
what technological trends are already evident ? and in which direction are they heading ?

We identified several trends:

- battery-powered laptops are now used for tasks previously performed using a mains-powered personal computer
- rechargeable handheld appliances are replacing mains-powered appliances
- hands-free devices are becoming a necessary extension of hand-held devices

The transition from mains powered to battery powered digital equipment is reflected in a number of microcontrollers we reviewed. We considered a range of processors designed for different purposes ranging from Motorola microcontrollers designed for automotive applications; to AVR microcontrollers used in a variety of household appliances; to the Ubicom microcontrollers used in mobile communications networks; to the ARM microcontrollers used in mobile handsets [1]. The increased memory capacity and reduced power supply requirements are shown in Figure 3; Figure 4 shows increasingly sophisticated on-chip peripherals and processor architecture; Harvard architecture – a design that uses an independent program bus and data bus – used in reduced instruction set computers (RISC) is introduced to conserve power. The increase in the processor speed is nevertheless accompanied by increased power consumption as shown in Figure 5.

Controller	Flash (bytes)	SRAM (bytes)	EEPROM (bytes)	ROM (bytes)	Pin-0uts	Supply Voltage (V)	Supply Current (mA)	Freq. Max (MHz)	No. of Power Saving Options
Motorola MC68HC908RF2	2k	128	-		12	1.8-3.3	4.3	4	2
Motorola MC68HC908JG16	16k	384	-		20	4-5.5	6.5	6	2
Motorola 68HC08KH12		384	-	12k	42	4-5.5	18	6	2
ATMega169 Butterfly	16k (+4Mbit data)	2k	512		32	2.7-5.5	10	16	5
ATMega16	16k	1k	512		32	2.7-5.5	12	16	6
Ubicom SX52BD/PQ	2k	262	2k		40	2.2-5.5	50	50	4
Motorola MC56F8323	32k (+8k data)	8k			46	2.25-2.75	60	60	4
ARM Processor AT91RM3400	128k	96k	128k		40	1.65-3.6	19	66	4

Figure 3.Memory capacities and electrical characteristics

Controller	Timers/PWM	Architecture	ADC	DAC	Coms
Motorola MC68HC908RF2	1 16-bit Timer	8-bit	-	-	
Motorola MC68HC908JG16	2 16-bit Timers	8-bit	1	-	SCI, USB
Motorola 68HC08KH12	2 16-bit Timer. 2 PWM Channels (16-bit)	8-bit	-	-	USB
ATMega169 (Butterfly)	2 8-bit Timers, 1 16-bit Timer. 4 PWM Channels (2 8-bit & 2 16-bit)	8-bit Harvard	1	-	USART, USI, SPI

ATMega16	2 8-bit Timers, 1 16-bit Timer. 2 PWM Channels (8 & 16-bit)	8-bit Harvard	1	-	USART, SPI
Ubicom SX52BD/PQ	2 16-bit Timers, 1 8-bit Timer. 2 PWM Channels (16-bit)	Harvard (8-bit data, 12-bit program)	1	-	Programmable Serial Interface controlled by Firmware Libraries
Motorola MC56F8323	8 16-bit Timers, 6 PWM Channels	32-bit	2	-	FlexCAN, 2 SCI, 2 SPI
ARM Processor AT91RM3400	16-bit System Timer, 2 16-bit Timers 2 PWM Channels (16-bit)	32-bit Harvard	-	-	4 USART, USB 2.0, MMX, 3 Synchronous Serial Controllers (32 bit), ISO7816 Smart Card, RS485, Modem Control, Infrared, SPI

Figure 4.Available on-chip peripherals.

3.1 New design

The preliminary prototype of the new jacket uses an ATMega16 microcontroller running at 16MHz [3]. The increased speed allows pulse width modulated (PWM) sine waves and other more complex wave forms to be produced.

It also allows different waveforms to be independently produced and distributed through different speakers. Multiplexing has increased the number of switches available. The ATMega16 also offers larger storage for applications associated with wave form manipulation and generation.

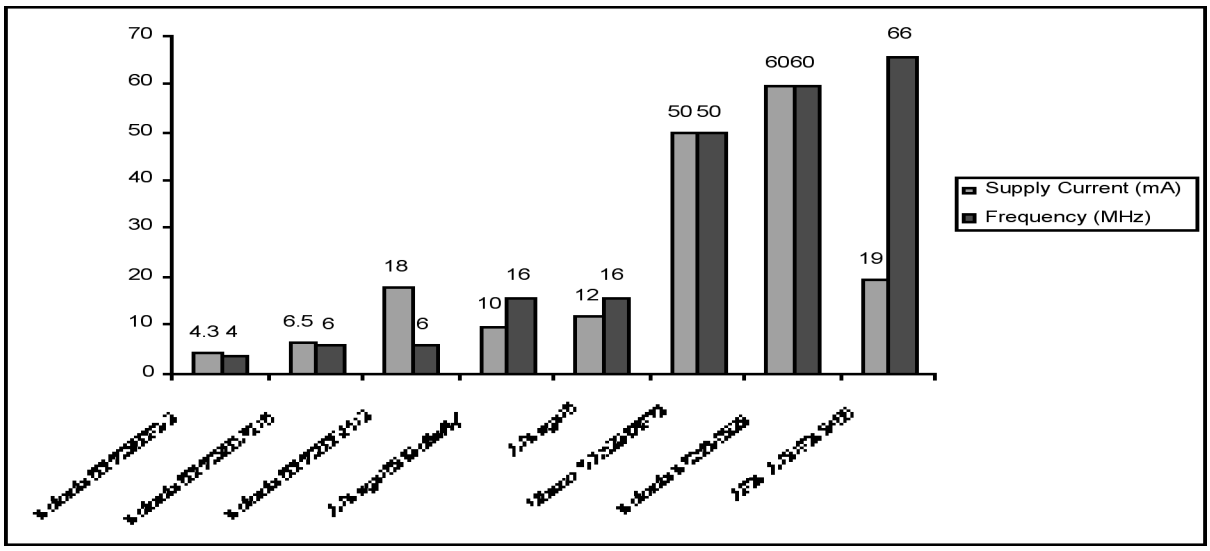


Figure 5.Current consumption versus frequency

3.2 Peripherals and communications

An important consideration for the new design is its user interface. Its communications capabilities must not compromise its mobility. Two microcontrollers are now being investigated.

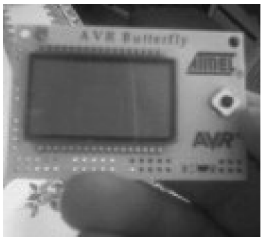


Figure 6.The AVR Butterfly chip

The AVR ATMega169, presented in a credit-card size badge called the AVR Butterfly, is powered by a 3-volt lithium battery [2]. It offers an assortment of peripherals ranging from LCD display, joystick, speaker, temperature sensor and light sensor. It also has USB capabilities, 4 Mbit of data flash and an RS232 level converter.

The Ubicom chip introduces the concept of virtual peripherals where software libraries provide the option of an SCI, SPI, Can bus, USB or Ethernet port. A new generation of this chip together with additional hardware allows implementation of 802.11 wireless LAN [20].

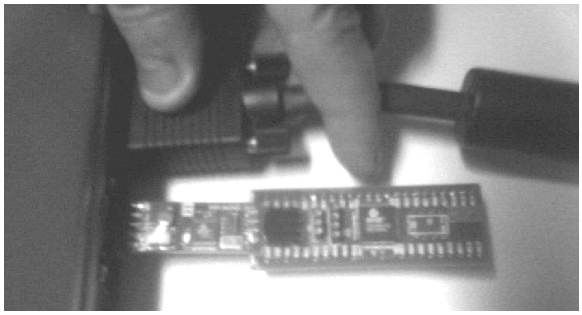


Figure 7. Ubicom SX56DB/PQ module with SX programming key plugged into RS-232 interface

3.3 E-field sensing

The Motorola e-field sensing chip (MC33794) offers new sensing capability. Though it was developed to monitor passenger safety in automotive air-bag controller design, in the context of the jacket it will provide a way to interpret the wearer's environment.

E-field sensing is closely related to one of the earliest electronic instruments, the Theremin. This chip originated in research from MIT Media Labs [15]. We used an MC33794 Evaluation Module based on a Motorola 68HC908QY4 microcontroller to test applications for controlling movement based on work begun by Phillippa Cullen in 1970.

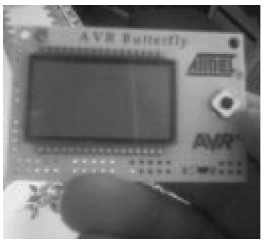


Figure 8 E-field evaluation module; MC33794 chip (top) reads up to 10 antennae.

A theremin signal traditionally monitors position of the body. One of our objectives was to derive velocity and acceleration as an alternative to using mercury tilt switches used in the jacket. This proved to be problematic due to limitations of the 68HC908QY4 used in this evaluation module [9]. An evaluation module based on Motorola's MC56F8300 offers a more sophisticated processor with an appropriate communications interface [10]. This approach to sensing will be pursued further once the MC56F8300 becomes available.



Figure 9.The Singing Jacket

3.4 Sound chips

The sound of the jacket can be enhanced using additional chips. Two chips have been reviewed: the Dream synthesiser and the Motorola Symphony chip. They communicate with a microcontroller via the SPI port.

The Dream Synthesiser (ATSAM9708) is a 128-voice wavetable synthesis chip [4]. Standard firmware includes equalisation, surround and an MPEG 2 audio decoder. The Dream chip includes the Roland GS Sound Set. Its MIDI ports communicate with a microcontroller via the SCI port. The chip's 3.3V supply voltage features a power-saving mode. It is well suited for mobile or wearable applications.

The Motorola Symphony chip (DSP56367) is a 24-bit Audio Digital Signal Processor [11]. The chip's DSP capabilities support current digital audio standards (Dolby, DVD Audio, 3D virtual surround). It operates with supply voltages between 1.8-3.3V, has two sleep modes and supports Enhanced Serial Audio Interface (ESAI) including I^2S, SPDIF, Sony and AC97.

The processing capabilities of the Symphony chip will be used to perform speaker compensation on the piezo-transducers. A precedent for this is the CSIRO's A4 chip frequency domain processor which convolve audio signal with the inverse of the frequency response [6].

4 Conclusion

The development of the jacket is seen as a possible prototype for an interactive musical instrument that will be a hands-free accessory for a mobile phone handset [17]. We think it is important for musicians

to anticipate such developments and not lose an opportunity to influence directions for the future.

5 Acknowledgements

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